

Summary and Recommendation

1. Objective

The primary objective of this analysis is to **understand customer churn, identify the customer segments associated with higher churn, and determine the major factors that appear to influence customer retention.**

The analysis covers **7,043 customers and 21 attributes**, including:

- Customer demographics
- Tenure
- Phone and internet services
- Additional services such as OnlineSecurity and TechSupport
- Contract type
- Payment method
- Monthly and total charges
- Churn status

The analysis uses data cleaning, descriptive statistics, and multiple visualizations such as **countplots, pie charts, histograms, and stacked percentage bar charts** to identify meaningful patterns.

2. Data Preparation & Quality

Before performing the analysis, the dataset was checked and cleaned:

- The dataset contains **7,043 records and 21 columns**.
- **TotalCharges** was initially stored as a string, so blank values were replaced with **0** and the column was converted into a numeric data type.
- After cleaning, the dataset contained **0 missing values**.
- The analysis also checked **customerID** for duplicates and found **0 duplicate customer IDs**.
- **SeniorCitizen** was converted from **0/1 into No/Yes** to make the visualizations easier to interpret.
- The average customer tenure is approximately **32.37 months**.
- Average monthly charges are approximately **\$64.76**.
- Average total charges are approximately **\$2,279.73**.

This preprocessing ensured that the subsequent analysis was based on a consistent and usable dataset.

3. Overall Churn Overview

The overall churn analysis is the starting point of the project.

Key Finding

- **26.54% of customers have churned.**
- **Approximately 73.46% of customers have remained.**

This means that roughly **1 out of every 4 customers** in the dataset has churned.

The churn distribution is therefore significant enough to require further investigation rather than treating churn as an isolated event. The subsequent charts were created specifically to understand **which customer characteristics and service combinations are associated with this churn.**

4. Demographic Analysis

Gender vs Churn

The gender-based countplot shows churn for male and female customers.

Insight

The difference between male and female customers is comparatively small. Therefore, **gender does not appear to be one of the strongest differentiating factors in this analysis.**

This is useful because it indicates that churn analysis should focus more heavily on **customer tenure, contracts, services, and payment behavior** rather than treating gender as a primary churn indicator.

5. Senior Citizen Analysis

The SeniorCitizen analysis provides a more noticeable difference.

Dataset composition

- Senior citizens represent approximately **16.21% of the customer base.**
- Non-senior citizens represent approximately **83.79%.**

The stacked percentage chart shows the **proportion of churned and retained customers within each SeniorCitizen category**.

Key Insight

A **comparatively greater percentage of senior citizens have churned** than non-senior citizens.

This makes SeniorCitizen status an important demographic dimension to consider when analyzing customer churn.

However, the chart represents an association in this dataset; it does not establish that age status itself causes churn.

6. Tenure Analysis

The tenure distribution was analyzed using a histogram with churn status.

Key Insights

The tenure chart shows a clear difference in the distribution of churned and retained customers:

- Churn is more concentrated among **customers with relatively short tenure**.
- As customer tenure increases, the distribution contains a larger proportion of customers who remain.
- Long-term customers therefore appear more represented among retained customers.

The dataset has:

- **Average tenure:** 32.37 months
- **Median tenure:** 29 months
- **Minimum tenure:** 0 months
- **Maximum tenure:** 72 months

Business Interpretation

The early period of the customer relationship appears particularly important. Customers who have only recently joined the service form a more visible portion of the churned population.

This suggests that **customer onboarding and early-stage engagement are important areas to monitor**.

7. Contract Analysis

The contract analysis compares:

- Month-to-month
- One year
- Two year

contracts against churn.

Key Insight

The chart shows a strong difference in churn distribution across contract types.

Month-to-month customers account for the largest share of churn, while customers with longer-term contracts show considerably fewer churned customers.

This indicates that contract structure is one of the more prominent patterns visible in the analysis.

Interpretation

Month-to-month customers may have fewer contractual commitments, meaning their relationship with the company can be more flexible. Longer-term customers appear more represented among retained customers.

Therefore, **contract type should be considered when segmenting customers for churn analysis**.

8. Internet Service Analysis

The service analysis examines:

- DSL
- Fiber optic
- No internet service

Key Insight

Fiber optic customers show a noticeably higher number of churned customers than DSL customers and customers without internet service.

This is particularly important because Fiber optic represents a major internet-service segment in the dataset.

The analysis therefore identifies **InternetService** as an important variable for further investigation, particularly the Fiber optic segment.

It would be useful in further analysis to examine whether this relationship is also connected with **MonthlyCharges, contract type, tenure, and additional support services**.

9. Additional Service Analysis

The analysis includes nine service-related variables:

- PhoneService
- MultipleLines
- InternetService
- OnlineSecurity
- OnlineBackup
- DeviceProtection
- TechSupport
- StreamingTV
- StreamingMovies

The 3×3 visualization makes it possible to compare these service categories systematically.

Important Pattern

For several additional services, particularly:

- **OnlineSecurity**
- **OnlineBackup**
- **DeviceProtection**
- **TechSupport**

customers **without these services show higher churn counts** than customers who have them.

Interpretation

Customers who do not use additional security, backup, device-protection, or technical-support services appear more frequently among churned customers.

This does **not prove that adding these services directly prevents churn**, but it identifies a meaningful relationship that deserves further investigation.

10. Payment Method Analysis

The payment-method chart compares churn across different payment methods.

The major payment categories include:

- Electronic check
- Mailed check
- Bank transfer
- Credit card

Key Insight

Electronic Check shows the highest churn presence among the payment-method categories in the analysis.

The notebook specifically identifies the pattern that a customer is more likely to churn when using **Electronic Check**.

This makes payment method an important variable for customer segmentation and further churn investigation.

Business Interpretation

The payment-method relationship could be investigated alongside:

- Contract type
- Tenure
- MonthlyCharges
- InternetService
- Customer services

because the observed churn pattern may reflect combinations of these variables rather than payment method alone.

11. Major Key Insights

Based on the charts and analysis performed in the notebook, the major findings are:

1. Overall churn is significant

26.54% of customers have churned, meaning approximately one-quarter of the customer base in the dataset has been lost.

2. Tenure is strongly associated with churn

Customers with **shorter tenure appear more frequently among churned customers**, while longer-tenure customers are more represented among retained customers.

3. Contract type shows a clear difference

Month-to-month customers have the strongest churn presence, while longer-term contract customers show substantially lower churn.

4. Fiber optic customers show higher churn

Among internet-service categories, **Fiber optic has a noticeably higher churn count** than DSL and customers without internet service.

5. Additional support/security services are associated with lower churn counts

Customers without **OnlineSecurity, OnlineBackup, DeviceProtection, and TechSupport** appear more frequently among churned customers.

6. Electronic Check is a prominent churn segment

The payment-method analysis shows **Electronic Check customers having the highest churn presence** among the payment categories.

7. Senior citizens show comparatively higher churn

Although senior citizens make up only **16.21% of the customer base**, their churn proportion is comparatively higher.

8. Gender shows comparatively less differentiation

The gender analysis does not show a major difference compared with the stronger patterns observed for contract, tenure, services, and payment method.

12. High-Risk Customer Profile Identified by the Analysis

The different charts can also be viewed together rather than independently.

The analysis identifies a customer profile that appears repeatedly across the higher-churn segments:

Short-tenure + Month-to-month contract + Fiber optic service + Electronic Check payment + absence of additional security/support services

This should **not be interpreted as a formal predictive model or proof that every customer with these characteristics will churn**. Instead, it represents a combination of characteristics that the exploratory analysis identifies as important areas for further investigation.

13. Business Implications

The analysis provides several areas that a business could monitor:

Customer onboarding

Since churn is more concentrated among shorter-tenure customers, the early customer lifecycle should receive particular attention.

Contract engagement

The large difference visible across contract types indicates that contract structure is an important dimension for customer-retention analysis.

Fiber optic segment

Because Fiber optic customers show comparatively higher churn, this segment could be analyzed further for issues involving pricing, service experience, support, or contract structure.

Additional services

The higher churn counts among customers without security and support services indicate that these service combinations should be examined further.

Payment behavior

Electronic Check customers represent an important churn segment and could be analyzed together with tenure, contract, and monthly charges.

14. Conclusion

The analysis demonstrates that **customer churn is not evenly distributed across the customer base**. The overall churn rate is **26.54%**, with the strongest visible patterns occurring around **tenure, contract type, internet service, payment method, SeniorCitizen status, and additional support/security services**.

The charts make these relationships easier to identify by combining **count-based analysis with percentage-based visualization**, particularly the SeniorCitizen stacked percentage chart. The analysis therefore provides a strong exploratory foundation for **customer segmentation, churn-risk monitoring, and further predictive analysis**.